Structural Variant Detection

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Introduction

Structural variants (SVs) – deletions, duplications, inversions, translocations, and complex rearrangements – are genomic events $\geq$ 50 bp. They often have larger phenotypic impact than single-nucleotide variants but are harder to call because they involve breakpoints across long distances. Short-read callers like Manta and Delly, plus long-read callers (Sniffles, pbsv), are standard.

Prerequisites

BAM files from high-coverage WGS; reference genome.

Theory

Short-read SV callers combine three signals: - **Read pairs** with abnormal insert size or orientation. - **Split reads** whose two halves map to distant loci. - **Read-depth** anomalies (CNV overlap).

Long-read callers directly observe breakpoint-spanning reads, dramatically improving resolution of repeat-rich SVs.

Assumptions

Sequencing depth is adequate ($\geq$ 30x WGS); insert-size distribution is approximately normal; reference matches the sample ancestry.

R Implementation

Manta/Delly run on the command line producing VCF; R ingests and annotates:

```r
library(VariantAnnotation)
library(StructuralVariantAnnotation)

# Use a VariantAnnotation-shipped VCF as a proxy
vcf_file <- system.file("extdata", "chr22.vcf.gz", package = "VariantAnnotation")
vcf <- readVcf(vcf_file, "hg19")
vcf_sv <- vcf[1:10, ]   # illustrative subset; treat as SV records

# breakpointRanges builds a GRanges of breakpoints from SV VCF INFO fields
gr <- breakpointRanges(vcf_sv)
length(gr)
head(as.data.frame(gr))
```

Example command-line calls:

```
manta configManta.py --bam sample.bam --reference ref.fa --runDir manta_out
delly call -g ref.fa -o sample.bcf sample.bam
```

Output & Results

A VCF of SV calls with type (DEL, DUP, INV, INS, BND), coordinates, and supporting-read counts; annotation against genes yields per-gene SV impact.

Interpretation

"WGS SV calling identified 4,120 events (2,800 deletions, 1,100 duplications, 180 inversions, 40 translocations) per sample. The germline 17q12 deletion spans RCAD and was concordant with the clinical suspicion."

Practical Tips

- Use **two** callers and intersect (high-precision) or union (high-sensitivity); any single caller misses events.
- Filter by breakpoint support; single-read evidence is almost always false.
- Long-read WGS resolves complex rearrangements that short reads cannot; use for structural-disease phenotyping.
- Tumour SV calling requires matched normal to remove germline events; somatic callers: Manta + DELLY (cancer mode), GRIDSS.
- Annotate breakpoints with `StructuralVariantAnnotation` to identify gene fusions and breakpoint-disrupted regions.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- RNA-seq with DESeq2 / edgeR
- Enrichment analysis (GSEA / ORA)
- scRNA-seq with Seurat